MCMC

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1. $(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)$ are iid samples

Linear Regression Model:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \quad i = 1, 2, \ldots, n$$

where β_0, β_1 are unknown parameters. X_i's are fixed covariates.

- ϵ_i's are (independent) uncorrelated.
- $X_i \perp\!\!\!\perp \epsilon_i$
- $E(\epsilon_i) = 0 \ \forall i$, $Var(\epsilon_i) = \sigma^2 \ \forall i$ $\Rightarrow \underset{n\times 1}{Y} = \underset{n\times 2}{X}\underset{2\times 1}{\beta} + \underset{n\times 1}{\epsilon}$
- $\epsilon_i \overset{iid}{\sim} N(0, \sigma^2)$, $i = 1, 2, \ldots, n$. $\sim N_n(X\beta, \sigma^2 I_n)$

$$\begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix} = \begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix}$$

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} = \frac{\sum_i (x_i - \bar{x}) y_i}{\sum_i (x_i - \bar{x})^2}$$

Q1. Consider conjugate priors on β_0, β_1, and σ^2. Then find the posterior dist^n.s.

$\pi(\beta_0 | Y, X)$, $\pi(\beta_1 | X, Y)$, $\pi(\sigma^2 | Y, X)$

Joint Post. Dist^n.s $\pi(\beta_0, \beta_1 | X, Y)$

$\pi(\underset{\sim}{\beta}) \equiv \underset{\sim}{\beta} = (\beta_0, \beta_1) \sim N_2(\mu_\beta, \Sigma_\beta)$

$\pi(\beta) = \pi(\beta_0)\pi(\beta_1) \equiv N(\mu_0, \sigma_0^2)\ N(\mu_1, \sigma_1^2)$

$$\pi(\beta_0, \beta_1, \sigma^2 | \underset{\sim}{Y}, \underset{\sim}{X}) \propto L(\beta_0, \beta_1, \sigma^2 | \underset{\sim}{Y}, \underset{\sim}{X})\ \pi(\beta_0, \beta_1, \sigma^2)$$

Prior: $\pi(\beta_0)\ \pi(\beta_1)\ \pi(\sigma^2) \equiv INV\text{-}G(\alpha_0, \lambda_0)$

$$\propto \frac{1}{(\sigma\sqrt{2\pi})^n} \exp\left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2 \right\} \pi(\beta_0)\ \pi(\beta_1)\ \pi(\sigma^2)$$

Case 1 $\underset{\sim}{\beta} = (\beta_0, \beta_1)^T \sim N_2(\mu_\beta, \Sigma_\beta)$

$$\pi(\underset{\sim}{\beta}, \sigma^2 | \underset{\sim}{Y}, \underset{\sim}{X}) \propto L(\underset{\sim}{\beta}, \sigma^2 | \underset{\sim}{Y}, \underset{\sim}{X})\ \pi(\underset{\sim}{\beta})\ \pi(\sigma^2)$$

$$\propto \frac{1}{(\sigma^2 \times 2\pi)^{n/2}} \exp\left\{ -\frac{1}{2\sigma^2} (\underset{\sim}{Y} - \underset{\sim}{X}\beta)^T (\underset{\sim}{Y} - \underset{\sim}{X}\beta) \right\} \times \frac{1}{2\pi\sqrt{|\Sigma_\beta|}} \exp\left\{ -\frac{1}{2} (\beta - \mu_\beta)^T \Sigma_\beta^{-1} (\beta - \mu_\beta) \right\}$$

$$\times \frac{\lambda_0^{\alpha_0}}{\Gamma(\alpha_0)} (\sigma^2)^{-\alpha_0 - 1} \exp\left\{ -\frac{\lambda_0}{\sigma^2} \right\}$$

$$\propto \exp\left\{ -\frac{1}{2} \left[\beta^T \left(\frac{X^T X}{\sigma^2} \right) \beta - 2\beta^T \frac{X^T Y}{\sigma^2} \right] - \frac{1}{2} \left[\beta^T \Sigma_\beta^{-1} \beta - 2\beta^T \Sigma_\beta^{-1} \mu_\beta \right] \right\}$$

$$(\sigma^2)^{-(\alpha_0 + \frac{n}{2} + 1)} \exp\left\{ -\frac{1}{\sigma^2} \left(\lambda_0 + \frac{Y^T Y}{2} \right) \right\}$$

$= \pi(\sigma^2 | Y, X)$

$$\propto \boxed{\exp\left\{ -\frac{1}{2} \left[\beta^T \left(\frac{X^T X}{\sigma^2} + \Sigma_\beta^{-1} \right) \beta - 2\beta^T \left(\frac{X^T Y}{\sigma^2} + \Sigma_\beta^{-1} \mu_\beta \right) \right] \right\}}_{= (1)} \times \pi(\sigma^2 | \underset{\sim}{Y}, \underset{\sim}{X})$$

$$\propto \pi(\underset{\sim}{\beta} | \underset{\sim}{Y}, \underset{\sim}{X}, \sigma^2)\ \pi(\sigma^2 | Y, X)$$

$$\beta^T \underbrace{\left(\frac{X^T X}{\sigma^2} + \Sigma_\beta^{-1} \right)}_{= G_{2\times 2}} \beta - 2\beta^T \underbrace{\left(\frac{X^T Y}{\sigma^2} + \Sigma_\beta^{-1} \mu_\beta \right)}_{= \underset{\sim}{g}_{2\times 1}}$$

$$= \beta^T G \beta - 2\beta^T g + g^T G^{-1} g - g^T G^{-1} g$$

$$= (\beta - G^{-1} g)^T G (\beta - G^{-1} g) - g^T G^{-1} g$$

$$= \beta^T G \beta - \beta^T G G^{-1} g - g^T G^{-1} G \beta + g^T G^{-1} G G^{-1} g - g^T G^{-1} g$$

$$\pi(\underset{\sim}{\beta} | \underset{\sim}{Y}, \underset{\sim}{X}, \sigma^2) \equiv N_2(G^{-1} \underset{\sim}{g}, G^{-1})$$

Gibbs Sampler

(1) $\pi(\underset{\sim}{\beta} | \underset{\sim}{Y}, \underset{\sim}{X}, \sigma^2) \equiv N_2(G^{-1} g, G^{-1})$

$$\pi(\underset{\sim}{\beta} | \underset{\sim}{Y}, \underset{\sim}{X}, \sigma^2) \propto \pi(\beta, \sigma^2, \underset{\sim}{Y}, \underset{\sim}{X})$$

$$\propto \pi(\beta, \sigma^2 | \underset{\sim}{Y}, \underset{\sim}{X})$$

$$\propto (1) \times \pi(\sigma^2 | \underset{\sim}{Y}, \underset{\sim}{X})$$

$$\propto \exp\left\{ -\frac{1}{2} (\beta^T G \beta - 2\beta^T g) \right\}$$

$$\equiv N_2(G^{-1} g, G^{-1})$$ ← Full Conditional distribution of $\underset{\sim}{\beta}$ given $\sigma^2, \underset{\sim}{Y}, \underset{\sim}{X}$.

(2) $\pi(\sigma^2 | \underset{\sim}{Y}, \underset{\sim}{X}, \underset{\sim}{\beta}) \sim \pi(\sigma^2 \times \ldots)$

$\equiv N_2\left(G^{-1}g, G^{-1}\right)$ ← Full Conditional distribution of β given $\sigma^2, \underline{Y}, \underline{X}$.

② $\pi(\sigma^2 \mid \underline{Y}, \underline{X}, \beta) \propto \pi(\beta, \sigma^2, \underline{Y}, \underline{X})$

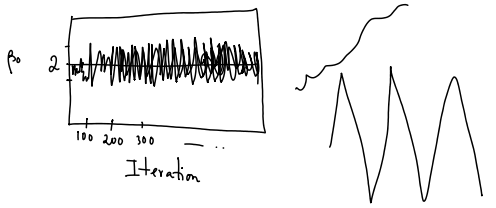
$\propto \pi(\beta, \sigma^2 \mid \underline{Y}, \underline{X})$

$\propto L(\beta, \sigma^2 \mid \underline{Y}, \underline{X})\, \pi(\sigma^2)$

$\propto \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^n \exp\left\{-\frac{1}{2\sigma^2}(Y-X\beta)^T(Y-X\beta)\right\} (\sigma^2)^{-(\alpha_0+1)} \exp\left\{-\frac{\lambda_0}{\sigma^2}\right\}$

$\propto (\sigma^2)^{-\left(\alpha_0+\frac{n}{2}+1\right)} \exp\left\{-\frac{1}{\sigma^2}\left(\lambda_0+\frac{1}{2}A\right)\right\}$ where $A = (Y-X\beta)^T(Y-X\beta)$

$\pi\left(\sigma^2 \mid \underline{Y}, \underline{X}, \beta\right) \equiv \text{Inv-}G\left(\alpha_0+\frac{n}{2},\ \lambda_0+\frac{1}{2}A\right)$ ← Full Conditional dist^n of σ^2 given Y, X, β.



H.W. Posterior dist^n of ① $\beta \mid \underline{Y}, \underline{X}$ ② $\sigma^2 \mid \underline{Y}, \underline{X}$ — Priors: $\pi(\beta) \equiv N_2(\mu_\beta, \Lambda_\beta)$, $\pi(\sigma^2) \equiv \text{Inv-}G(\alpha_0, \lambda_0)$

$\pi(\beta \mid \underline{Y}, \underline{X}) = \int_0^\infty \pi(\beta, \sigma^2 \mid \underline{Y}, \underline{X})\, d\sigma^2$

$\pi(\sigma^2 \mid \underline{Y}, \underline{X}) = \int_{\mathbb{R}^2} \pi(\beta, \sigma^2 \mid \underline{Y}, \underline{X})\, d\beta$

[H.W. $Y_i = \beta_1 X_i + \epsilon_i$; $\epsilon_i \sim N(0, \sigma^2)$

Prior : $\pi(\beta_1) \equiv N(\tau_1, s_1^2)$, $\pi(\sigma^2) \equiv \text{Inv-}G(\alpha_0, \lambda_0)$

→ ① Find the posterior dist^{n}s of β_1 and σ^2.

② Find the full conditional dist^{n}s of β_1 & σ^2.

$\underline{Y} = X\beta + \underline{\epsilon}$, $\underline{\epsilon} \sim N_n(X\beta, \sigma^2 I_n)$

$\pi(\beta, \sigma^2 \mid \underline{Y}, X) \propto L(\beta, \sigma^2 \mid \underline{Y}, X)\, \pi(\beta, \sigma^2 \mid X)$ → $\equiv \pi(\beta \mid X)\, \pi(\sigma^2 \mid Y, X)$

$\propto (\sigma^2)^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\sigma^2}(Y-X\beta)^T(Y-X\beta)\right\} \exp\left\{-\frac{1}{2}(\beta-\mu_\beta)^T\Sigma_\beta^{-1}(\beta-\mu_\beta)\right\} (\sigma^2)^{-(\alpha_0+1)} \exp\left\{-\frac{\lambda_0}{\sigma^2}\right\}$

$\equiv N_2(\mu_\beta, \Sigma_\beta)\ \text{Inv-}G(\alpha_0, \lambda_0)$

$= \underbrace{\pi(\beta \mid \sigma^2, Y, X)}_{\text{Conditional posterior dist}^n \text{ of } \beta \text{ given } \sigma^2}\ \underbrace{\pi(\sigma^2 \mid Y, X)}_{\text{Marginal Posterior dist}^n \text{ of } \sigma^2}$

① Find $\pi(\beta \mid Y, X)$ = Marginal Posterior dist^n of β given Y, X.

⊙ $E(\beta \mid Y, X)$, $\text{Var}(\beta \mid Y, X)$ ← $\pi(\beta \mid Y, \underline{X})$

⊙ $E(\beta \mid \sigma^2, Y, X)$, $\text{Var}(\beta \mid \sigma^2, Y, X)$ ← $\pi(\beta \mid \sigma^2, Y, X)$

⊙ $E(\sigma^2 \mid Y, X)$, $\text{Var}(\sigma^2 \mid Y, X)$ ← $\pi(\sigma^2 \mid Y, X)$

⊙ $E(\sigma^2 \mid \beta, Y, X)$, $\text{Var}(\sigma^2 \mid \beta, Y, X)$ ← $\pi(\sigma^2 \mid \beta, Y, X)$

① Let $\theta \mid \lambda \sim N\left(\mu, \frac{\tau^2}{\lambda}\right)$ and $\lambda \sim \text{Gamma}\left(\frac{1}{2}, \frac{1}{2}\right)$ (shape, rate)

Find the marginal dist^n of θ.

$\pi(\theta) = \int_0^\infty \pi(\theta \mid \lambda)\, \pi(\lambda)\, d\lambda = \frac{\tau}{\pi\left\{\tau^2 + (\theta-\mu)^2\right\}}$; $\theta \in \mathbb{R}$, $\mu \in \mathbb{R}$, $\tau > 0$

$\theta \sim \text{Cauchy}(\mu, \tau)$.

② Let X be a single observation and $X \mid \theta \sim N(\theta, \sigma^2)$ where σ^2 is known.

Assume a prior of θ as $\pi(\theta) \equiv \text{Cauchy}(\mu, \tau)$ where μ, τ are hyperparameters

(2) Let X be a single observation and $X|\theta \sim N(\theta, \sigma^2)$ where σ^2 is known.

Assume a prior of θ as $\pi(\theta) \equiv \text{Cauchy}(\mu, \tau)$, where μ, τ are hyperparameters.

How do you simulate θ from its posterior distribution?

$$\pi(\theta|x) \propto L(\theta|x)\,\pi(\theta) \propto \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{1}{2\sigma^2}(x-\theta)^2\right\} \times \frac{1}{\pi\left\{\tau^2+(\theta-\mu)^2\right\}}$$

$$\propto \frac{\exp\left\{-\frac{1}{2\sigma^2}(x-\theta)^2\right\}}{\left\{\tau^2+(\theta-\mu)^2\right\}}$$

The posterior distn. of θ has not any closed-form expression. So, direct simulation is not possible.

[θ_{prop} Acceptance prob $\left(\theta_{prop}, \theta^{(t)}\right)$

$\theta^{(t+1)} = \theta_{prop}$ w.p. $p\left(\theta^{(t)}, \theta_{prop}\right)$

$= \theta^{(t)}$ w.p. $1 - p\left(\theta^{(t)}, \theta_{prop}\right)$] $\longrightarrow$ MH algorithm.

Gibbs sampler

$$\pi(\theta, \lambda|x) \propto \pi(\theta, \lambda, x) \propto \pi(x|\theta)\,\pi(\theta|\lambda)\,\pi(\lambda)$$

$$\propto \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{1}{2\sigma^2}(x-\theta)^2\right\} \times \frac{\sqrt{\lambda}}{\tau\sqrt{2\pi}} \exp\left\{-\frac{\lambda}{2\tau^2}(\theta-\mu)^2\right\} \times \frac{1}{\sqrt{2}\,\Gamma\left(\frac{1}{2}\right)} \exp\left\{-\frac{\lambda}{2}\right\} \lambda^{\frac{1}{2}-1}$$

(1) $$\pi(\lambda|\theta, x) \propto \pi(\theta, \lambda, x) \propto \pi(x|\theta)\,\pi(\theta|\lambda)\,\pi(\lambda)$$

$$\propto \pi(\theta|\lambda)\,\pi(\lambda)$$

$$\propto \exp\left\{-\frac{\lambda}{2\tau^2}\left[\tau^2+(\theta-\mu)^2\right]\right\}$$

$\lambda|\theta, x \sim$ Exponential distn. with rate parameter $\dfrac{\tau^2+(\theta-\mu)^2}{2\tau^2}$

(2) $$\pi(\theta|\lambda, x) \propto \pi(\theta, \lambda, x) \propto \pi(x|\theta)\,\pi(\theta|\lambda)\,\pi(\lambda)$$

$$\propto \pi(x|\theta)\,\pi(\theta|\lambda)$$

$$\equiv N\left(\frac{\tau^2}{\tau^2+\lambda\sigma^2}x + \frac{\lambda\sigma^2}{\tau^2+\lambda\sigma^2}\mu,\ \frac{\tau^2\sigma^2}{\tau^2+\lambda\sigma^2}\right)$$

Step-1 Initiate $\theta = \theta^{(0)}$

Step-2 $\lambda^{(0)} \sim \pi\left(\lambda|\theta^{(0)}, x\right) \equiv \text{Exp}\left(\dfrac{\tau^2+(\theta^{(0)}-\mu)^2}{2\tau^2}\right)$

Step-3 $\theta^{(1)} \sim \pi\left(\theta|\lambda^{(0)}, x\right) \equiv$

Step-4 $\lambda^{(1)} \sim \pi\left(\lambda|\theta^{(1)}, x\right) \equiv$

$\lambda^{(t)} \sim \pi\left(\lambda|\theta^{(t)}, x\right)$

$\theta^{(t+1)} \sim \pi\left(\theta|\lambda^{(t)}, x\right)$

$$\lambda^{(t+1)} \sim \pi(\lambda \mid \theta^{(t)}, x)$$
$$\theta^{(t+1)} \sim \pi(\theta \mid \lambda^{(t)}, x)$$

Step-n : Repeat until convergence.

(3) Let a single observation $(X_1, X_2) \sim N_2\left(\begin{pmatrix}\mu_1\\ \mu_2\end{pmatrix}, \Sigma = \begin{pmatrix}\sigma_{11} & \sigma_{12}\\ \sigma_{21} & \sigma_{22}\end{pmatrix}\right)$

Suppose Σ is known. You have to generate $\begin{pmatrix}\mu_1\\ \mu_2\end{pmatrix}$.

$\sigma_{11} = \sigma_1^2$
$\sigma_{22} = \sigma_2^2$
$\sigma_{12} = \rho\,\sigma_1\sigma_2$

Choose prior $\pi(\mu_1, \mu_2) \propto 1$.

Generate $\begin{pmatrix}\mu_1\\ \mu_2\end{pmatrix}$ using the Gibbs sampling algorithm.

① $\pi(\mu_1 \mid \mu_2, X_1, X_2) \equiv N\left(X_1 + \rho\frac{\sigma_1}{\sigma_2}(\mu_2 - X_2),\ \sigma_1^2(1-\rho^2)\right)$

② $\pi(\mu_2 \mid \mu_1, X_1, X_2) \equiv N\left(X_2 + \rho\frac{\sigma_2}{\sigma_1}(\mu_1 - X_1),\ \sigma_2^2(1-\rho^2)\right)$

Check -??

$\pi(\mu_1, \mu_2 \mid X_1, X_2) \equiv N_2\left(\begin{pmatrix}X_1\\ X_2\end{pmatrix}, \Sigma\right)$

(4) $(X_{1i}, X_{2i}) \overset{iid}{\sim} N_2\left(\begin{pmatrix}\mu_1\\ \mu_2\end{pmatrix}, \Sigma\right)$, $i = 1, 2, \ldots, n$,

and Σ is known.

$\underline{X}_1 = \begin{pmatrix}X_{11}\\ \vdots\\ X_{1n}\end{pmatrix}$, $\underline{X}_2 = \begin{pmatrix}X_{21}\\ \vdots\\ X_{2n}\end{pmatrix}$

① $\pi(\mu_1 \mid \mu_2, \underline{X}_1, \underline{X}_2)$
② $\pi(\mu_2 \mid \mu_1, \underline{X}_1, \underline{X}_2)$

$\longrightarrow$ Perform Gibbs sampling algorithm.